MAT1190 Lecture Notes

Arky!! :3c

’26 Fall Semester

Contents

Day 1: Introduction (Sep. 8, 2026)

This is the first part of a two-semester course. Throughout this class, we will cover Hartshorne Chapter II. To quote Hunter Spink, "if I just went through Hartshorne straight, it would not make any sense; this textbook does not make any sense. As a first exposure, the book is not reasonable. " We will get through scheme theory, though. The goal of this class is to develop scheme theory as a main topic, then show how it applies to a bunch of easier theories. "A lot of people think schemes are so confusing, that you basically have to do 6 to 9 on a completely obsolete theory to show you how it actually matters" (but he won't do that).

Three references that are very useful, outside of Hartshorne.

(i) Ravi Vakil's *The Rising Sea*. This is meant to be a yearlong textbook.

(ii) Jo Harris' *The Geometry of Schemes*. Harris just says "this is schemes, suck it up" then develops a bunch of properties about it.

(iii) Qin Li's *Arithmetic* (something something).

The class will have an *oral* midterm and final (roughly October and December). There will be ungraded exercises (biweekly, as we go?) to turn in throughout the semester; the orals will be based on the easier questions.

This class will start with the theory of manifolds over $\mathbb{R}$, i.e., the classical theory of nice spaces. Consider the following insight on the Gelfand–Kolmogorov theorem: consider $C^0(M,\mathbb{R})$, the ring of continuous functions $M \to \mathbb{R}$ on a manifold M. How do we recover M from $C^0(M,\mathbb{R})$? Colloquially, different points of M see different aspects of any function in $C^0(M,\mathbb{R})$. If we start with a point $p \in M$, we get an $\mathbb{R}$-algebra map $C^0(M,\mathbb{R}) \to \mathbb{R}$ given by evaluating the functions at the given point p; we call this the "evaluation map", denoted

$$\mathrm{ev}_p \colon C^0(M,\mathbb{R}) \to \mathbb{R}, \quad f \mapsto f(p).$$

Specifically, $\mathrm{ev}_p \in \mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(C^0(M,\mathbb{R}),\mathbb{R})$. What does it mean for ev_p to be an $\mathbb{R}$-algebra map? As an example, while $\mathrm{Hom}(\mathbb{C},\mathbb{C}) = \mathrm{Aut}(\mathbb{C})$ contains "uncountably many horrible things", we see that $\mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(\mathbb{C},\mathbb{C})$ consists of homomorphisms that fix $\mathbb{R}$, i.e., it only admits the identity and complex conjugation.

Claim 1.1. $\{\mathrm{ev}_p\}_{p\in M}$ are the only elements of $\mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(C^0(M,\mathbb{R}),\mathbb{R})$.

We will do the proof for $M = \mathbb{R}$, then see that the proof follows analogously for several other examples of M as well.

Proof. Assume we have an $\mathbb{R}$-algebra map $\Phi \colon C^0(M,\mathbb{R}) \to \mathbb{R}$ that sends the identity function x to a, i.e., $x \mapsto a$. We claim that this map is ev_a; if not, then there is some function $g \in C^0(M,\mathbb{R})$ such that $g \not\mapsto g(a)$ under Φ. Indeed, consider $\ker(\Phi)$; we see that $x - a \in \mathrm{Ker}(\Phi)$; now, consider the function

$$(x-a)^2 + (g(x)-b)^2,$$

where $b \coloneqq \Phi(g(x))$. This function was constructed to never vanish; the only way for it to be zero would be for $x = a$ and $g(x) = \Phi(g(x))$, but this never happens, so it is nonzero everywhere. However, $x - a \in \ker\Phi$ and $g(x) - b \in \ker\Phi$, so

$$(x-a)^2 + (g(x)-b)^2 \in \ker\phi,$$

whence it is a unit; this would mean

$$1 = \frac{1}{(x-a)^2 + (g(x)-b)^2} \cdot \left((x-a)^2 + (g(x)-b)^2\right) \in \ker\Phi,$$

which is contradictory. □

Following Grothendieck, let R be any ring, and consider Spec R, which is a set of points with a topology, and is a locally ringed space (i.e., local functions "glue" to form global functions). We will rigorously define Spec R later. The key insight about manifolds is that, since they consist of charts that are compatible to each other locally, it allows us to "glue" data from functions together. Another important thing to know about locally ringed spaces is that if a function is non-vanishing everywhere, then it is invertible. Together, this lets you recover a topology on your ring R. Note that this stuff is still intentionally hand-wavey for now.[1]

Consider the polynomials over the real numbers, $\mathbb{R}[x]$. It is a ring. What would happen if we tried to reconstruct the space according to the procedure described earlier? Let's look at the $\mathbb{R}$-algebra homomorphisms from $\mathbb{R}[x] \to \mathbb{R}$; then we get that it is $\mathbb{R}$, since $x \mapsto a \in \mathbb{R}$, and there is probably some way to proceed from here.

There is, however, a reason we don't do algebraic geometry over the real numbers; that reason being, the process would suggest that $\mathbb{R}[x]$ is that ring of functions on the real line. If we looked at $x^2 + 1 \in \mathbb{R}[x]$, we would see that it's a polynomial that doesn't vanish anywhere and not invertible. This makes these "space-recovery" arguments impossible, specifically the part where a "non-vanishing function is invertible". There are two solutions to this behavior... either you invert functions which don't vanish everywhere, so you map $\mathbb{R} \to \mathbb{R} \setminus \{0\}$, or you can add points for every possible evaluation to every field. For example, consider

$$\mathrm{ev}_i : \mathbb{R}[x] \to \mathbb{C}, \quad x \mapsto i;$$

we would see that $x^2 + 1 \mapsto 0$. Next time, we will define that Spec(R) is built out of all homomorphisms from $\mathbb{R}$ to fields, modded out by "some stuff".

[1] "you... uncover something so unbelievably terrible that i can't even begin to describe it. something that's going to ruin your life for the next... twelve weeks!"

Day 2: Spectrum of a Ring (Sep. 10, 2026)

Last class, we talked about evaluation maps. Let M be a manifold, and let $p \in M$. We have that all of the $\mathbb{R}$-algebra maps $C^0(M, \mathbb{R}) \to \mathbb{R}$ were given by $\mathrm{ev}_p \colon C^0(M, \mathbb{R}) \to \mathbb{R}$. A key fact about locally ringed spaces is that any function on them which is nonvanishing everywhere is invertible; when working with polynomials $\mathbb{R}[x]$, though, we see that $x^2 + 1$ constitutes a counterexample to the aforementioned statement.

The function $x^2 + 1$ maps $\mathbb{R}$ to $\mathbb{R} \setminus \{0\}$. Observe that we may identify $xy - 1 = 0$ with $\mathbb{R} \setminus \{0\}$ by the isomorphism $(x, y) \mapsto x$; the curve $xy - 1 = 0$ (visualized, perhaps, in the xy-plane) admits an automorphism $(x, y) \mapsto (y, x)$; if we pre-compose $(x, y) \mapsto x$ with this, we see that we get an automorphism on $\mathbb{R} \setminus \{0\}$ in the form of $x \mapsto x^{-1}$. This is called the *Rabinowitsch trick*.

Our solution to the above problem is to "add more points". Consider $\mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(\mathbb{R}[x], \mathbb{C})$ (note that we extended the codomain $\mathbb{R}$ to $\mathbb{C}$). Fix a field K, and consider $f_1, \ldots, f_r \in K[x_1, \ldots, x_n]$, and consider the set

$$\{f_1(x_1, \ldots, x_n) = f_r(x_1, \ldots, x_n) = 0\} \subset K^n.$$

This is the zero set $z_K(f_1, \ldots, f_r)$.

Theorem 2.1. We have the canonical bijection,

$$\mathrm{Hom}_{K-\mathrm{alg}}\left(\frac{K[x_1, \ldots, x_n]}{(f_1, \ldots, f_r)}, K\right) \cong Z_K(f_1, \ldots, f_r).$$

Proof. The key fact about polynomial rings is that, in general, for any ring R, $\mathrm{Hom}_{R-\mathrm{alg}}(R[x_1, \ldots, x_n], S)$ is in a very natural bijection with n-tuples of elements of S. Concretely, you can send $(x_1, \ldots, x_n) \mapsto (s_1, \ldots, s_n)$ because $x_1, \ldots, x_n$ don't have to satisfy any relation over R. In this manner, any map from the quotient ring to K (in the original theorem statement) must send $f_1, \ldots, f_r$ to 0. Per our key fact, it follows that $f \mapsto \mathrm{ev}_{(s_1, \ldots, s_n)} = f(s_1, \ldots, s_n)$.

The canonical bijection thus arises for $(a_1, \ldots, a_n) \in Z_K(f_1, \ldots, f_r)$ and its associated evaluation maps $\mathrm{ev}_{(a_1, \ldots, a_n)}$. $\square$

Thus, we want to consider $K[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$ to be the "ring of functions". We want to ask, what is the function f_1 on the space $Z_k(f_1, \ldots, f_r)$? Clearly, $f_1|_{Z_k(f_1, \ldots, f_r)} = 0$ (and the same holds for each of $f_1, \ldots, f_r$).

As an example, consider a field K, $f_1 = x$, and $f_2 = y$. Then $Z_K(f_1, f_2) = (0, 0)$, and $K[x, y]/(f_1, f_2) \cong K$. If $f_1 = x$ an d $f_2 = y^2$, then

$$\frac{K[x, y]}{(x, y^2)} \cong \frac{K[y]}{y^2}.$$

Here, $\mathrm{ev}_{(0,0)}(y) = 0$, but we aren't setting y to 0 (as we are quotienting out by y^2). Thus, we have a hurdle to deal with; (not all) functions will not be determined by their points.

Let L/K be a field over K, and consider $Z_L(f_1, \ldots, f_r)$, where $f_1, \ldots, f_r$ are polynomials over K. Since K is embedded in L, we can consider $f_1, \ldots, f_r$ as polynomials over L as well, and write

$$Z_L(f_1, \ldots, f_r) = \{(a_1, \ldots, a_n) \mid f_1(a_1, \ldots, a_n) = \cdots = f_r(a_1, \ldots, a_n) = 0\} \subset L^n.$$

He then drew a diagram.

Theorem 2.2. (i) $\mathrm{Hom}_{K-\mathrm{alg}}(K[x_1,\ldots,x_n]/(f_1,\ldots,f_r),L)$ is in bijection with $Z_L(f_1,\ldots,f_r)$.

(ii) Let $R = K[x_1,\ldots,x_n]/(f_1,\ldots,f_r)$. We have that

$$\mathrm{Hom}_{K-\mathrm{alg}}(R,L) \cong \{(\text{prime ideals } p \subset R), \kappa(p) \xhookrightarrow{K-\mathrm{alg}} L)\},$$

where $\kappa(p) = \mathrm{Frac}(R/p)$.

Basically, this theorem says that "instead of being really good at solving system of equations, you could be really good at finding prime ideals instead."

Proof. For (i), the proof is the same as earlier. For (ii), we have a K-algebra map $\Phi\colon R \to L$; because the image is a field, we can factor the map through intermediate rings canonically. Because $\mathrm{Im}\,\Phi$ is an integral domain, Φ factors through

$$R \to R/p \xhookrightarrow{K-\mathrm{alg}} L,$$

where $\mathrm{Im}\,\Phi = R/p$. In particular, since L is a field, we can further write

$$R \to R/p \hookrightarrow \mathrm{Frac}(R/p) \xhookrightarrow{K-\mathrm{alg}} L.$$

The first two maps are canonical to the prime ideal p. □

"To solve a system of equations over a larger field, we specify a prime ideal, then specify an embedding."

Definition 2.3. We define $|\mathrm{Spec}(R)| \coloneqq \{p \subset R \mid p \text{ prime}\}$.

Corollary 2.4. If R is a field, then $|\mathrm{Spec}(R)|$ is a single point.

Of all the points that constitute $\mathrm{Spec}(R)$, of which they are prime ideals, some of them are $\ker \mathrm{ev}_p$ for $p \in Z_K(f_1,\ldots,f_r)$.

As an example, consider $|\mathrm{Spec}\,\mathbb{R}[x]|$. $\mathbb{R}[x]$ is a UFD, so its prime ideals are of the form (f) with f irreducible. Specifically, $|\mathrm{Spec}\,\mathbb{R}[x]| = \{(x-a) \mid a \in \mathbb{R}\} \cup \{(x-a)^2 + b^2 \mid a \in \mathbb{R}, b \sim -b \in \mathbb{R}^*\}$ (i.e., we identify b with $-b$ because it doesn't matter here). Following our process from earlier, we have

$$\mathbb{R}[x] \to \mathrm{Frac}\left(\frac{\mathbb{R}[x]}{(x-a)}\right) \xhookrightarrow{\mathbb{R}-\mathrm{alg}} L,$$

where $L/\mathbb{R}$. Here, $\mathrm{Frac}(\mathbb{R}[x]/(x-a)) \cong \mathbb{R}$. For the irreducible quadratics, we have

$$\mathbb{R}[x] \to \mathrm{Frac}\left(\frac{\mathbb{R}[x]}{((x-a)^2+b^2)}\right) \xhookrightarrow{\mathbb{R}-\mathrm{alg}} L,$$

where $\mathrm{Frac}(\mathbb{R}[x]/((x-a)^2+b^2)) \cong \mathbb{C}$. Thus, $|\mathrm{Spec}\,\mathbb{R}[x]|$ is given by $\mathbb{C}$ modded out by complex conjugation, with $\{0\}$ added in.

Now, consider

$$\mathbb{R}[x] \to \mathrm{Frac}(\mathbb{R}[x]/(0)) \hookrightarrow L,$$

where X satisfies no algebraic equation over $\mathbb{R}$.